

Elevation in Obesity: Uncovering the Geographical Aspects of Health

ECON 5293

Machine Learning with Causal Inference

Tyler Ransom

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Abstract

Geographic and environmental factors show strong correlations with obesity rates across the United States. This study examines the relationship between elevation, environmental contamination, and obesity using county-level data from 2010–2022. Initial county-level results indicate that a 1,000-foot increase in elevation predicts approximately 0.3-0.8 fewer obese individuals per 100 people, even after controlling for demographic, socioeconomic, and environmental factors. I also find evidence linking oil and gas extraction activities to higher obesity rates.

1 Introduction

The prevalence of obesity in the United States varies considerably across geographic regions, with some areas experiencing significantly higher rates than others. These disparities suggest that factors beyond individual choices and behaviors may contribute to obesity outcomes. This research explores the potential relationship between geographic and environmental factors—particularly elevation—and county-level obesity rates.

1.1 Background and Motivation

Obesity is a major public health concern in the United States, affecting approximately 30.9% of the population at the county level according to our data. While nutritional choices, physical activity, and genetic factors are well-documented contributors to obesity, emerging research suggests that geographic and environmental factors may also play significant roles. Understanding these relationships could inform more targeted public health interventions and provide insights into the complex determinants of obesity.

Prior research has identified potential physiological effects of altitude on metabolism, with some studies suggesting that higher elevations may be associated with lower obesity rates. Additionally, environmental quality factors such as the presence of extractive industries (mines, oil wells, gas wells) might indirectly affect health outcomes through various pathways including air quality, water contamination, or socioeconomic changes in communities. This study aims to quantify these relationships using county-level data across the United States.

1.2 Literature Review

Geographic variation in obesity rates has been documented in numerous epidemiological studies. Researchers have found significant disparities in obesity prevalence across states, counties, and even neighborhoods ([Ogden et al., 2014](#)). Several studies have examined

the relationship between elevation and health outcomes, including obesity.

The relationship between elevation and obesity has been consistently documented in the literature. [Voss et al. \(2013\)](#) and [Voss et al. \(2014\)](#) found that Americans living at higher altitudes are less likely to be obese than those living closer to sea level, controlling for several demographic and environmental factors. They proposed that hypoxia conditions at higher elevations might affect appetite hormones and metabolic processes. Similarly, [Woolcott et al. \(2014\)](#) observed that residing at higher altitudes is associated with lower obesity rates among individuals living in the United States. This negative correlation between bodyweight and elevation has been further supported by studies from [Díaz-Gutiérrez et al. \(2016\)](#), [Woolcott et al. \(2016\)](#), and [Merrill \(2020\)](#).

There is also a substantial literature in health economics demonstrating that health outcomes, including obesity, are determined in part by an individual’s immediate environment ([Chyn and Katz, 2021](#); [Deryugina and Molitor, 2021](#); [Ding, 2021](#); [Datar, Nicosia, and Samek, 2022](#)). This literature provides a foundation for examining how geographic and environmental factors might influence obesity rates.

Regarding environmental quality, [Hendryx and Ahern \(2008\)](#) examined the relationship between mining activities and public health outcomes, finding associations between mining presence and various adverse health outcomes in Appalachian communities. Limited research has explored the specific relationship between extractive industries and obesity rates, although mechanisms such as environmental stress, infrastructure limitations affecting physical activity, and socioeconomic changes could plausibly link these factors.

Recent studies have attempted to establish causal relationships between environmental factors and health outcomes using mover designs. [Hut and Oster \(2022\)](#) showed that diet is largely unresponsive to major household events, including a change of address, at least in the short term. Works by [Hut \(2020\)](#) and [Allcott et al. \(2020\)](#) demonstrated that household diet is largely unresponsive to changes in local environment in the short run, which presents methodological challenges for establishing causal relationships.

Huang, Liu, and Zuppann (2026) used a mover design to examine how non-cognitive skills in youth respond to relative body weight differences, providing a potential methodological approach for this study.

Despite this growing body of research, there remains a gap in understanding the causal relationship between environmental contamination and obesity, particularly regarding natural resource extraction activities. To my knowledge, this study represents the first examination of the relationship between obesity and natural resource extraction activities, as well as the first to incorporate hydrology data into a geographic analysis of obesity. Understanding these relationships contributes to the broader literature on environmental determinants of health and could inform place-based health interventions.

1.3 Research Question

This study addresses the following research questions:

1. Is there a significant relationship between county elevation and obesity rates in the United States?
2. Do environmental factors such as the presence of mines, oil wells, and gas wells correlate with county-level obesity rates?
3. To what extent do these relationships persist after controlling for socioeconomic, demographic, and regional factors?

Understanding these relationships contributes to the broader literature on environmental determinants of health and could inform place-based health interventions.

2 Data

2.1 Data Sources

For this project, I used a comprehensive county-level dataset from a public repository containing information on obesity rates, geographic features, environmental quality indicators, and sociodemographic characteristics. The original dataset contained 166,579 observations across 128 variables, spanning multiple years. After cleaning and filtering for complete data on key variables, my primary analytical sample includes 37,156 county-year observations from 2009 to 2022.

The obesity rate data likely originates from the CDC's Behavioral Risk Factor Surveillance System (BRFSS), which provides county-level estimates of obesity prevalence. Elevation data appears to be derived from geographic information systems, while information on extractive industries (mines, oil wells, gas wells) likely comes from federal or state regulatory agencies that track these facilities.

2.2 Variable Descriptions

Table 1: Description of Variables

Variable	Description
obesity_rate	Percentage of adults with BMI ≥ 30 in a county
elev000	County elevation in thousands of feet above sea level
has_mines	Binary indicator for presence of active mines in a county (1 = yes, 0 = no)
has_oil	Binary indicator for presence of oil wells in a county (1 = yes, 0 = no)
has_gas	Binary indicator for presence of gas wells in a county (1 = yes, 0 = no)
pct_bach_deg	Percentage of county population with a bachelor's de- gree
pct_grad_deg	Percentage of county population with a graduate degree
pct_walk	Percentage of county population who walk to work
log_inc	Natural logarithm of median household income in a county
median_age	Median age of county population
coastal	Binary indicator for coastal county (1 = yes, 0 = no)
dem_shr	Democratic vote share in most recent presidential elec- tion
pm2_5	Fine particulate matter (PM2.5) concentration in county air
smoker_rate	Percentage of adults who smoke in a county

Source: Public county-level dataset, 2009-2022.

2.3 Summary Statistics

Table 2: Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Max
obesity_rate	37,156	30.92	4.74	10.70	58.90
elev000	37,156	1.39	1.62	-0.06	11.78
has_mines	37,156	0.22	0.41	0	1
has_oil	37,156	0.30	0.46	0	1
has_gas	37,156	0.28	0.45	0	1
pct_bach_deg	37,156	9.25	3.81	0	36.57
log_inc	37,156	10.77	0.26	9.85	11.96
median_age	37,156	40.89	5.23	21.40	68.10

Note: This table presents summary statistics for key variables in the primary analytical sample (Sample 1, 2009-2022).

Table 3: Obesity Rate by Elevation Quartile

Elevation Quartile	Count	Mean	Std. Dev.	Min	Max
1 (Lowest)	9,351	32.32	5.30	14.6	53.7
2	9,260	31.47	3.86	15.3	57.7
3	9,372	31.37	3.81	18.9	58.9
4 (Highest)	9,173	28.47	4.88	10.7	52.5

Note: This table shows the distribution of obesity rates across elevation quartiles, revealing a clear pattern of lower obesity rates in higher-elevation counties.

2.4 Missing Data

The data cleaning process involved removing observations with missing values in key variables, which reduced the initial dataset from 166,579 to 37,156 observations (a reduction of 77.7%). I established two analytical samples: Sample 1 containing observations from

2009-2022 with complete data on core variables ($n = 37,156$), and Sample 2 comprising a subset with additional air quality, smoking rates, and weather variables ($n = 20,203$).

To ensure analytical consistency across models, I maintained the same approach to missing data throughout. Rather than imputing missing values, which might introduce bias given the geographic and temporal patterns in the data, I opted for complete-case analysis. While this approach reduces sample size, it maintains data integrity for the variables of interest. The substantial size of the remaining sample provides sufficient statistical power for the analyses.

3 Methodology

3.1 Empirical Framework

To examine the relationship between elevation and obesity rates, I estimate a series of regression models with increasing levels of control. The baseline specification is:

$$\text{Obesity}_{ct} = \beta_0 + \beta_1 \text{Elevation}_c + \varepsilon_{ct} \quad (1)$$

where Obesity_{ct} is the obesity rate in county c at time t , Elevation_c is the elevation of county c (in thousands of feet), and ε_{ct} is the error term.

I then augment this model with controls for environmental factors, socioeconomic characteristics, and demographic variables:

$$\text{Obesity}_{ct} = \beta_0 + \beta_1 \text{Elevation}_c + \beta_2 \text{Env}_{ct} + \beta_3 \text{SES}_{ct} + \beta_4 \text{Demo}_{ct} + \varepsilon_{ct} \quad (2)$$

where Env_{ct} includes environmental factors (presence of mines, oil wells, gas wells), SES_{ct} includes socioeconomic status indicators (education levels, income), and Demo_{ct}

includes demographic characteristics (median age, racial composition).

3.2 Estimation Strategy

I employ ordinary least squares (OLS) regression with robust standard errors clustered at the county level to account for serial correlation in the error terms. To address potential omitted variable bias from unobserved regional characteristics, I estimate models with various fixed effects specifications:

$$\text{Obesity}_{ct} = \beta_0 + \beta_1 \text{Elevation}_c + \beta_2 \text{Env}_{ct} + \beta_3 \text{SES}_{ct} + \beta_4 \text{Demo}_{ct} + \alpha_s + \varepsilon_{ct} \quad (3)$$

where α_s represents state fixed effects, controlling for time-invariant characteristics at the state level.

Additionally, I estimate models with watershed region fixed effects and state-by-watershed region fixed effects to control for geographic characteristics that might correlate with both elevation and health outcomes:

$$\text{Obesity}_{ct} = \beta_0 + \beta_1 \text{Elevation}_c + \beta_2 \text{Env}_{ct} + \beta_3 \text{SES}_{ct} + \beta_4 \text{Demo}_{ct} + \alpha_s + \gamma_w + \delta_{sw} + \varepsilon_{ct} \quad (4)$$

where γ_w represents watershed region fixed effects and δ_{sw} represents state-by-watershed region interaction fixed effects.

3.3 Robustness Checks

To assess the robustness of the findings, I estimate the models on two different samples: Sample 1 includes observations from 2009-2022 with complete data on core variables, while Sample 2 includes additional controls for air quality, smoking rates, and weather

variables. Consistency across these samples would strengthen confidence in the results.

I also examine the relationship between elevation and obesity by elevation quartile to explore potential nonlinearities. The descriptive statistics by quartile provide initial evidence of a gradient effect, with obesity rates decreasing as elevation increases.

4 Results

4.1 Main Findings

Table 4 presents the main regression results examining the relationship between elevation and obesity rates. Across all specifications, elevation shows a statistically significant negative association with county-level obesity rates.

Table 4: Regression Results - Effect of Elevation on Obesity Rates

	Model Specification				
	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Sample 1 (2009-2022)</i>					
Elevation (1000 ft)	-1.210***	-1.105***	-0.519***	-0.901***	-0.418***
	(0.044)	(0.036)	(0.049)	(0.049)	(0.050)
R^2	0.171	0.544	0.660	0.592	0.668
Observations	37,156	37,156	37,156	37,156	37,156
<i>Panel B: Sample 2 (With Additional Controls)</i>					
Elevation (1000 ft)	-1.240***	-1.074***	-0.519***	-0.977***	-0.428***
	(0.046)	(0.038)	(0.049)	(0.049)	(0.051)
R^2	0.198	0.583	0.705	0.627	0.713
Observations	20,203	20,203	20,203	20,203	20,203
Basic Controls		✓	✓	✓	✓
State FE			✓		✓
Watershed Region FE				✓	✓
Additional Controls (Sample 2)		✓	✓	✓	✓

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors clustered at the county level are in parentheses. Basic controls include binary indicators for mines, oil wells, and gas wells, education levels (bachelor's and graduate degrees), log of median income, median age, coastal status, Democratic vote share, and year fixed effects. Additional controls in Sample 2 include air quality (PM2.5), smoking rates, and summer temperature and precipitation.

Table 5 shows the coefficients for key control variables from the model with controls (without fixed effects).

Table 5: Coefficients for Key Control Variables

Variable	Coefficient	Standard Error
has_mines	-0.137	0.113
has_oil	1.288***	0.211
pct_bach_deg	-0.220***	0.023
log_inc	-2.375***	0.278

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Results from the model with controls (without fixed effects) using Sample 1.

4.2 Interpretation

The results consistently demonstrate a statistically significant negative relationship between elevation and county-level obesity rates. In the bivariate model, a 1,000-foot increase in elevation is associated with a 1.21 percentage point decrease in the obesity rate. After controlling for various socioeconomic, demographic, and environmental factors, this effect attenuates slightly to 1.105 percentage points.

Adding state fixed effects substantially reduces the magnitude of the elevation coefficient to -0.519, suggesting that some of the elevation effect is captured by state-level factors. Similarly, including watershed region fixed effects attenuates the coefficient to -0.901. In the most demanding specification with state-by-watershed region fixed effects, a 1,000-foot increase in elevation is associated with a 0.418 percentage point decrease in obesity rates.

The similar results across Sample 1 and Sample 2 indicate that the findings are robust to the inclusion of additional controls for air quality, smoking rates, and weather variables. The R-squared values increase substantially from the bivariate model (0.171) to the most comprehensive model (0.668), indicating that the control variables and fixed effects explain a considerable portion of the variation in obesity rates.

Among the control variables, several show significant associations with obesity rates. The presence of oil wells is associated with a 1.288 percentage point increase in obesity rates, although the presence of mines and gas wells does not show statistically significant associations. Higher educational attainment (as measured by the percentage of county residents with bachelor’s degrees) and higher median income are both significantly associated with lower obesity rates, consistent with established socioeconomic gradients in health outcomes.

4.3 Additional Results

The descriptive statistics by elevation quartile (Table 3) reveal a clear pattern: counties in the highest elevation quartile have a mean obesity rate of 28.47%, nearly 4 percentage points lower than counties in the lowest elevation quartile (32.32%). This gradient persists across quartiles, with obesity rates gradually decreasing as elevation increases.

The correlation between elevation and obesity rate is -0.414, indicating a moderate negative relationship. This correlation, combined with the regression results, provides strong evidence of an inverse relationship between elevation and obesity rates that persists after controlling for numerous potential confounding factors.

5 Conclusion

5.1 Summary of Findings

This study investigated the relationship between geographic and environmental factors—particularly elevation—and county-level obesity rates. The analyses consistently demonstrated a statistically significant negative association between elevation and obesity, with a 1,000-foot increase in elevation associated with a 0.418 to 1.240 percentage point decrease in obesity rates, depending on the model specification. This relationship remained robust across various control variables and fixed effects specifications.

Additionally, the presence of oil wells showed a significant positive association with obesity rates, while higher educational attainment and income were associated with lower obesity rates. These findings contribute to the understanding of how geographic and environmental factors may influence health outcomes beyond individual behaviors and choices.

5.2 Implications

These findings have several implications for public health policy and research. First, they suggest that geographic and environmental factors should be considered in obesity prevention efforts. Place-based interventions might need to be tailored to geographic contexts, with different approaches potentially needed in low-elevation versus high-elevation regions.

Second, the association between extractive industries (particularly oil wells) and higher obesity rates suggests a need for closer examination of how these industries affect community health. This could inform environmental justice initiatives and regulatory policies for extractive industries.

Third, the strong socioeconomic gradients in obesity rates underscore the importance of addressing education and income inequalities as part of comprehensive obesity prevention strategies.

5.3 Limitations and Future Research

This study has several limitations. First, while the analysis controls for many potential confounders, the possibility of omitted variable bias remains. Second, the cross-sectional nature of the analysis limits causal interpretation; counties with lower obesity rates might differ from other counties in unobserved ways that correlate with elevation but are not captured in the models.

Future research could explore potential mechanisms linking elevation to obesity

rates, such as physiological effects of altitude on metabolism, differences in physical activity patterns due to geographic features, or cultural and dietary differences in high-elevation communities. Additionally, longitudinal studies tracking changes in obesity rates following environmental changes (e.g., new extractive industry developments) could provide stronger evidence for causal relationships.

Research could also investigate potential nonlinearities in the elevation-obesity relationship and examine whether this relationship varies across different regions, demographic groups, or time periods. Finally, qualitative research could complement these quantitative findings by exploring how residents in different geographic contexts perceive and navigate their environments in ways that might influence health behaviors and outcomes.

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A Additional Tables and Figures

Table 6: Correlation Matrix for Key Variables

	obesity_rate	elev000
obesity_rate	1.000	-0.414
elev000	-0.414	1.000

Note: This table presents the correlation between obesity rate and elevation (in thousands of feet), showing a moderate negative relationship.

B Technical Details

B.1 Data Cleaning Process

The data cleaning process involved the following steps:

1. Removing observations with missing values in key variables (obesity rate, elevation, mines, oil wells, gas wells)

2. Selecting 27 relevant variables from the original 128
3. Standardizing data types for consistent analysis
4. Creating transformed variables:
 - Binary indicators for mines, oil wells, and gas wells
 - Log transformations for heavily skewed variables (oil wells, gas wells, income, population)
 - Elevation quartiles to examine nonlinear relationships
5. Sorting data by county and year for panel analysis
6. Creating analytical samples based on data completeness